2.1 Answers

October 20, 2023

1

(c): Product of Powers Property.
(b): Negative Exponent Property.
(a): Power of a Product Property.

2

Scientific notation is defined to be a number in the form $c \times 10^n$ where $1 \leq c \leq 10$ and n is an integer. The number 25.2×10^{-3} is such that $c = 25.2 > 10$, thus meaning the given number is not in scientific notation.

3

$3^3 \times 3^2 = 3^{3+2} = 3^5$ (Prodcut of Powers Property)

4

$(4^{-2})^3 = 4^{-2\times 3} = 4^{-6}$ (Power of a Power Property)
$= \frac{1}{4^6}$ (Negative Exponent Property)

5

$(-5)(-5)^4 = (-5)^1 \times (-5)^4 = (-5)^5$ (Product of Powers Property)

6

$(2^4)^2 = 2^{4\times 2} = 2^8$ (Power of a Power Property)

7

$\frac{5^2}{5^5} = 5^{2-5} = 5^-3$ (Quotient of Powers Property)
$= \frac{1}{5^3}$ (Negative Exponent Property)

8

$(\frac{3}{5})^4 = \frac{3^4}{5^4} = \frac{81}{625}$ (Power of a Quotient Property)

9

$(\frac{2}{7})^{-3} = \frac{2^{-3}}{7^{-3}}$ (Power of a Quotient Property)
$= \frac{\frac{1}{8}}{\frac{1}{343}}$ (Negative Exponent Property)
$= \frac{343}{8}$

10

$9^3 \times 9^{-1} = 9^{3-1} = 9^2$ (Product of Powers Property)

11

$\frac{3^4}{3^{-2}} = 3^{4+2} = 3^6$ (Quotient of Powers Property)

12

$(\frac{2}{3})^{-5}(\frac{2}{3})^4 = (\frac{2}{3})^{-5+4} = (\frac{2}{3})^{-1}$ (Product of Powers Property)
$= \frac{1}{\frac{2}{3}} = \frac{3}{2}$ (Negative Exponents Property)

13

$6^3 \times 6^0 \times 6^{-5} = 6^{3+0} \times 6^{-5} = 6^3 \times 6^{-5} = 6^{3-5} = 6^-2$ (Product of Powers Property)
$= \frac{1}{36}$ (Negative Exponents Property)

14

$((\frac{1}{2})^{-5})^2 = (\frac{1}{2})^{-5\times2} = (\frac{1}{2})^{-10}$ (Power of a Power Property)
$= \frac{1}{(\frac{1}{2})^{10}}$ (Negative Exponents Property)
$= \frac{1}{\frac{1^{10}}{2^{10}}} = \frac{1}{\frac{1}{1024}} = 1024$ (Power of a Quotien Property)

15

$(4.2 \times 10^3)(1.5 \times 10^6) = (4.2 \times 1.5)(10^3 \times 10^6) = 6.3 \times 10^9$

16

$(1.2 \times 10^{-3})(6.7 \times 10^{-7}) = (1.2 \times 6.7)(10^{-3} \times 10^{-7}) = 8.04 \times 10^{-10}$

17

$(6.3 \times 10^5)(8.9 \times 10^{-12}) = (6.3 \times 8.9)(10^5 \times 10^{-12}) = 56.07 \times 10^{-7} = 5.607 \times 10^1 \times 10^{-7} = 5.607 \times 10^{-6}$

18

$(7.2 \times 10^9)(9.4 \times 10^8) = (7.2 \times 9.4)(10^9 \times 10^8) = 67.68 \times 10^{17} = 6.768 \times 10^{18}$

19

$(2.1 \times 10^{-4})^3 = 2.1^3 \times (10^{-4})^3 = 9.261 \times 10^{-12}$

20

$(4.0 \times 10^3)^4 = 4.0^4 \times (10^3)^4 = 256 \times 10^12 = 2.56 \times 10^{14}$

21

$\frac{8.1 \times 10^{12}}{5.4 \times 10^9} = \frac{8.1}{5.4} \times \frac{10^{12}}{10^9} = 1.5 \times 10^3$

22

$\frac{1.1 \times 10^{-3}}{5.5 \times 10^{-8}} = \frac{1.1}{5.5} \times \frac{10^{-3}}{10^{-8}} = 0.2 \times 10^5 = 2 \times 10^4$

23

$\frac{(7.5 \times 10^8)(4.5 \times 10^{-4})}{1.5 \times 10^7} = \frac{(7.5 \times 4.5)(10^8 \times 10^{-4})}{1.5 \times 10^7}$
$\rightarrow = \frac{3.375 \times 10^5}{1.5 \times 10^7} = 2.25 \times 10^{-2}$

24

$\frac{w^{-2}}{w^6} = w^{-2-6} = w^{-8}$ (Quotient of Powers Property)
$= \frac{1}{w^8}$ (Negative Exponents Property)

25

$(2^2y^3)^5 = 4^5 \times (y^3)^5$ (Power of a Product Property)
$= 1024y^{15}$ (Power of a Power property)

26

$(p^3q^2)^{-1} = (p^3)^{-1} \times (q^2)^{-1}$ (Power of a Product Property)
$= p^{-3}q^{-2}$ (Power of a Power Property)
$= \frac{1}{p^3q^2}$ (Negative Exponents Property)

27

$(w^3x^{-2})(w^6x^{-1}) = (w^3 \times w^6)(x^{-2} \times x^{-1}) = w^9x^{-3}$ (Product of Powers Property)

$= \frac{w^9}{x^3}$ (Negative Exponents Property)

28

$(5s^{-2}t^4)^{-3} = (5s^{-2})^{-3} \times (t^4)^{-3}$ (Power of a Product Property)
$= 5^{-3}s^6t^{-12}$ (Power of a Product Property, Power of a Power Property)
$= \frac{s^6}{125t^{12}}$ (Negative Exponents Property)

29

$(3a^3b^5)^{-3} = (3a^3)^{-3} \times (b^5)^{-3}$ Power of a Product Property)
$= 3^{-3}a^{-9}b^{-15}$ (Power of a Product Property, Power of a Power Property)

$= \frac{1}{27a^9b^{15}}$ (Negative Exponents Property)

30

$\frac{x^{-1}y^2}{x^2y^{-1}} = x^{-1-2}y^{2+1} = x^{-3}y^3$ (Quotient of a Power Property)
$= \frac{y^3}{x^3}$ (Negative Exponent Property)

31

$\frac{3c^3d}{9cd^{-1}} = \frac{c^2d^2}{3}$ (Quotient of Powers Property)

32

$\frac{4r^4s^5}{24r^4s^{-5}} = \frac{r^0s^{10}}{6} = \frac{s^{10}}{6}$ (Quotient of Powers Property)

33

$\frac{2a^3b^{-4}}{3a^5b^{-2}} = \frac{2a^{-2}b^{-2}}{3}$ (Quotient of Powers Property)
$= \frac{2}{3a^2b^2}$ (Negative Exponents Property)

34

$$\begin{array}{ll} & \frac{y^{11}}{4z^3} \times \frac{8z^7}{y^7} \\ \rightarrow & \frac{8z^7y^{11}}{4z^3y^7} \\ \rightarrow & 2z^4y^4 \textit{ Quotient of Powers Property} \end{array}$$

35

$$\begin{array}{ll} & \frac{x^2y^{-3}}{3y^2} \times \frac{y^2}{x^{-4}} \\ \rightarrow & \frac{x^2y^{-3}y^2}{x^{-4}3y^2} \\ \rightarrow & \frac{x^2y^{-1}}{x^{-4}3y^2} \ Product\ of\ Powers\ Property \\ \rightarrow & \frac{x^6y^{-3}}{3} \ Quotient\ of\ Powers\ Property \\ \rightarrow & \frac{x^6}{3y^3} \ Negative\ Exponents\ Property \end{array}$$

36

B

37

The error here is that the powers of the denominator and numerator were divided. The Quotient of Powers Property states that the powers should in fact be subtracted, meaning the correct answer is $x^{10-2} = x^8$

38

The error here is that the powers of the factors of the product were multiplied. The Product of Powers Property actually states that the powers should be added, not multiplied, meaning the correct answer is $x^{5+3} = x^8$.

39

The error here is that (-3) and (-3) were multiplied together. However, there is no properrty that $a^m b^n = (ab)^{m+n}$ where $m \neq n$, meaning this conclusion is incorrect. Instead, we can use the Product of Powers property to conclude that $(-3)^2(-3)^4 = (-3)^6$

40

We are given that $A = \frac{\sqrt{3}}{4}s^2$. With the given value for s, we can find A using this formula:

$$\begin{aligned} & A = \tfrac{\sqrt{3}}{4}(\tfrac{x}{3})^2 \\ \rightarrow \quad & \tfrac{\sqrt{3}}{4}\tfrac{x^2}{9} \\ \rightarrow \quad & \tfrac{x^2\sqrt{3}}{36} \end{aligned}$$

41

$$\begin{aligned} & V = \pi r^2 h \\ \rightarrow \quad & \pi x^2 \tfrac{x}{2} \\ \rightarrow \quad & \tfrac{\pi x^3}{2} \end{aligned}$$

42

$$\begin{aligned} & V = lwh \\ \rightarrow \quad & \tfrac{5x}{3} 2x \times x \\ \rightarrow \quad & \tfrac{10x^3}{3} \end{aligned}$$

43

$$x^{15}y^{12}z^8 = x^4y^7z^{11} \times x^{11}y^5z^{-3}$$

44

$$3x^3y^2 = \tfrac{12x^2y^5}{4x^{-1}y^3}$$

45

$$(a^5b^4)^2 = a^{14}b^{-1} \times a^4b^9$$

46

$$x^{12}y^{16} = (x^8y^8)(x^4y^8) = (x^6y^9)(x^6y^7) = (x^2y^{10})(x^{10}y^6)$$

47

$$\begin{aligned} & a^{-n} = \tfrac{a^0}{a^n} \ \textit{Quotient of Powers Property} \\ \rightarrow \quad & \tfrac{a^0}{a^n} = \tfrac{1}{a^n} \ \textit{Zero Exponenet Property} \\ \rightarrow \quad & a^{-n} = \tfrac{1}{a^n} \ \textit{Transitive Property of Equality} \end{aligned}$$

48

$$\begin{aligned} & (\tfrac{a}{b})^n = (a \times \tfrac{1}{b})^n = a^n(\tfrac{1}{b})^n \ \textit{Product of Powers Property} \\ \rightarrow \quad & a^n(\tfrac{1}{b})^n = a^n(b^{-1})^n \ \textit{Negative Exponents Property} \\ \rightarrow \quad & a^n(b^{-1})^n = a^nb^{-n} \ \textit{Power of a Power Property} \\ \rightarrow \quad & a^nb^{-n} = \tfrac{a^n}{b^n} \ \textit{Negative Exponents Property} \\ \rightarrow \quad & (\tfrac{a}{b})^n = \tfrac{a^n}{b^n} \ \textit{Transitive Property} \end{aligned}$$

49

Volume of Pacific $= (1.56 \times 10^{14})(4.03 \times 10^3) = (1.56 \times 4.03)(10^{14} \times 10^3) = 6.2868 \times 10^{17}$
Volume of Atlantic $= (7.68 \times 10^{13})(3.93 \times 10^3) = (7.68 \times 3.93)(10^{13} \times 10^3) = 30.1824 \times 10^{16} = 3.01824 \times 10^{17}$
Volume of Indian $= (6.86 \times 10^{13})(3.96 \times 10^3) = (6.86 \times 3.96)(10^{13} \times 10^3) = 27.1656 \times 10^{16} = 2.71656 \times 10^{17}$
Volume of Arctic $= (1.41 \times 10^{13})(1.21 \times 10^3) = (1.41 \times 1.21)(10^{13} \times 10^3) = 1.7061 \times 10^{16}$

50

We are given that the South American continent has been moving about 0.000022 miles per year for the past $125,000,000$ years, meaning the total amount that the continent has moved in that time is $0.000022 \times 125,000,000$. This can be

simplified further into scientific notation:

$$\begin{aligned} & 0.000022 \times 125,000,000 \\ \rightarrow \quad & 0.000022 \times 1.25 \times 10^8 \\ \rightarrow \quad & 0.0000275 \times 10^8 = 2.75 \times 10^3 \end{aligned}$$

51

We are comparing the volumes of a bead and a pearl, both of which we will assume are in spherical shapes, meaning their volume can be given by: $\frac{4}{3}\pi r^3$. The bead has a diameter of 6 millimeters, or a radius of 3 millimeters, while the pearl has a radius of 4.5 millimeters, meaning we can compare their volumes:

$$\rightarrow \quad \frac{\frac{4}{3}\pi 3^3}{\frac{4}{3}\pi 4.5^3} \\ \frac{27}{91.125} = \frac{8}{27}$$

meaning the ratio of the volume of the bead to the volume of the pearl is $\frac{8}{27}$

52

(a): Because we are considering three spheres, the total volume of the three tennis balls is $3(\frac{4}{3}\pi r^3) = 4\pi r^3$
(b): The volume of a cylinder (and this cylinder) is $\pi r^2 h$
(c): We know that the diameter of each of the tennis balls is $2r$. With the information that h is equal to the sum of the three diameters of the tennis balls, we can conlude that $h = 2r + 2r + 2r = 6r$
(d): Now that we have an expression of h in terms of r, we can rewrite the volume of the can, and find how much of it is taken up by the three tennis balls:

$$\rightarrow \quad \frac{4\pi r^3}{\frac{2}{3}}$$

meaning the balls take up $\frac{2}{3}$ of the can.

53

(a): $V = \pi 9.53^2 \times 1.55 = \pi \times 140.772395\ Cubic\ millimeters = 1.40772395\pi \times 10^{-11}$ cubic meter.
(b): ?
(c): ?

54

(a): We are given that the radius of earth is 5 times that of the inner core, which has a radius of r, meaning the radius of earth is $5r$, and its volume is $\frac{4}{3}\pi(5r)^3 = \frac{4}{3}\pi 125r^3$. The volume of the inner core is $\frac{4}{3}\pi r^3$, meaning the volume of Earth's total volume to the volume of the inner core is $\frac{\frac{4}{3}\pi 125r^3}{\frac{4}{3}\pi r^3} = 125$

(b): We are given that the radius of the outer core is $\frac{11}{6}r + r = \frac{17}{6}r$, meaning the volume of the outer core is $\frac{4}{3}\pi(\frac{17}{6}r)^3 = \frac{4}{3}\pi\frac{4913}{216}r^3$, meaning the ratio of the volume of Earth's outer core to the volume the inner core is: $\frac{\frac{4}{3}\pi\frac{4913}{216}r^3}{\frac{4}{3}\pi r^3} = \frac{4913}{216} \approx 22.7$